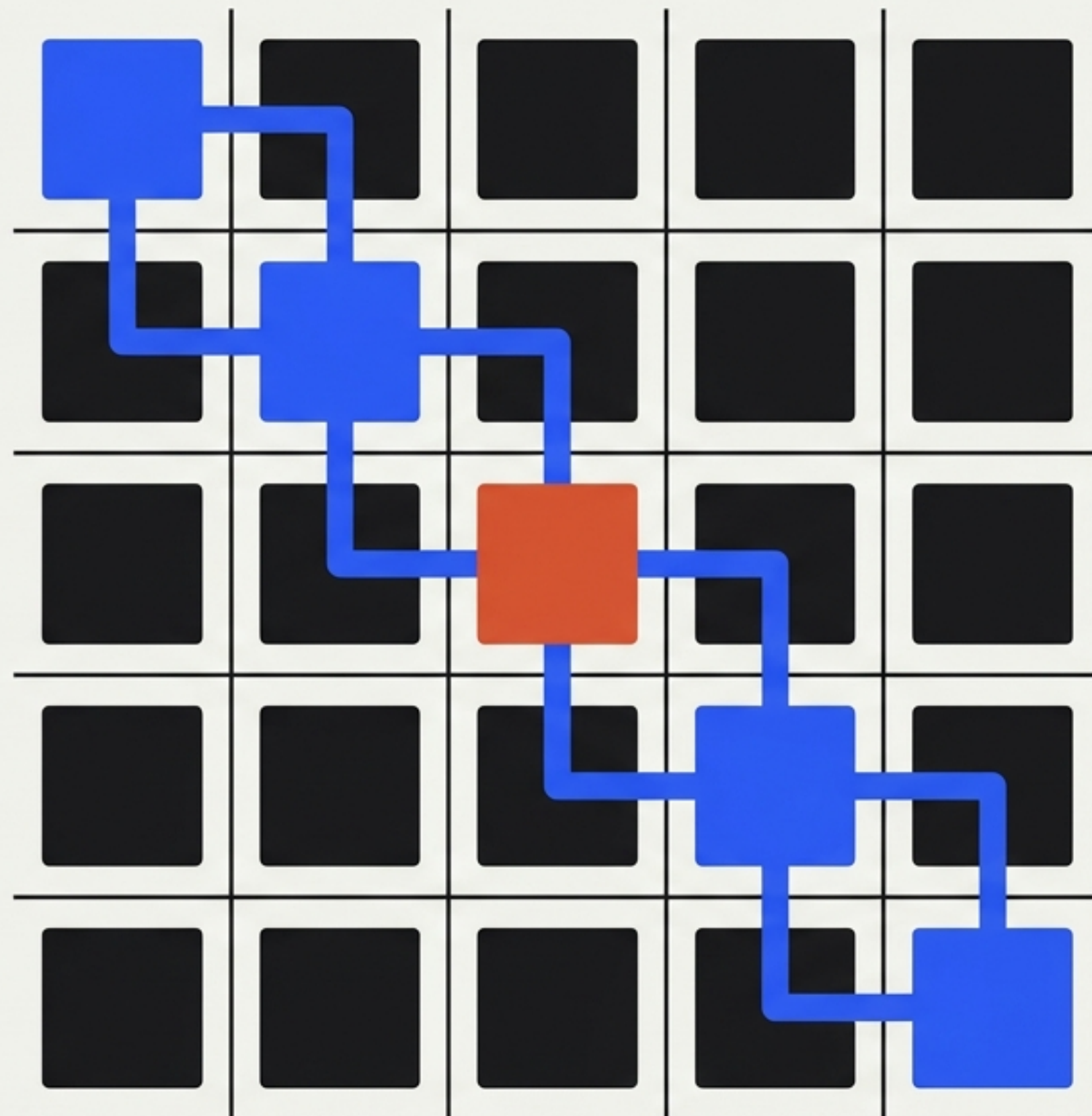


Mastering Missing Data with MICE

A Visual Algorithmic Walkthrough of Multivariate Imputation by Chained Equations.

Guide: Algorithmic Imputation /
Level: Intermediate



Not all missing data is created equal

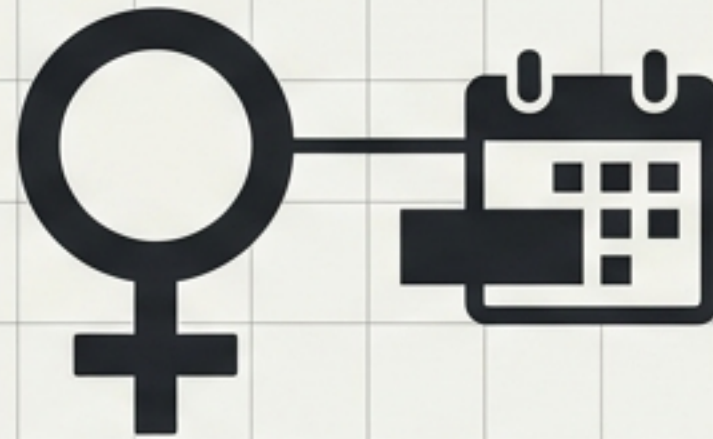
MCAR



Data is lost with no pattern.

Example: A survey lost in the mail. The missingness is unrelated to any variable.

MAR



TARGET USE CASE FOR MICE

Missingness depends on other observed variables.

Example: A specific demographic hesitating to reveal their age.

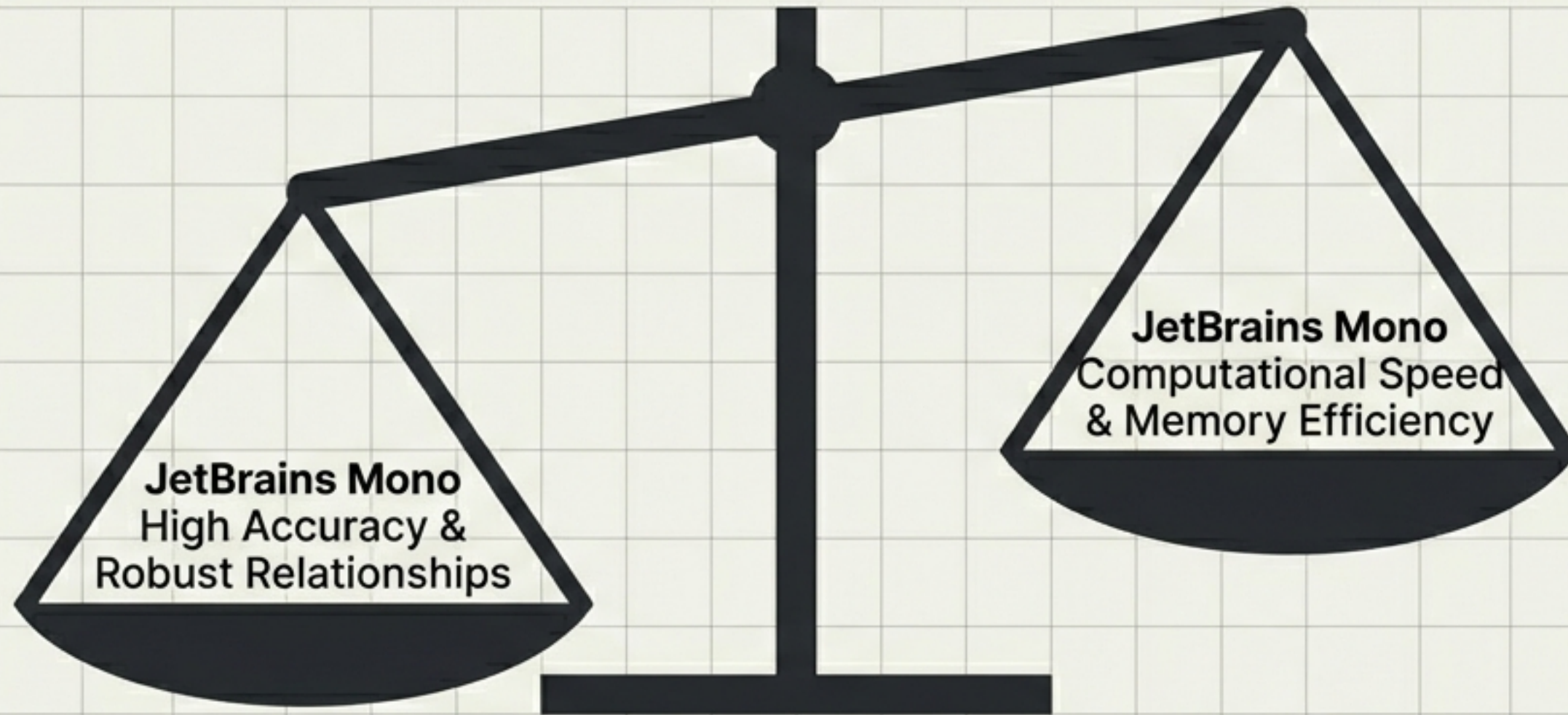
MNAR



Deliberate omission.

The missingness is related to the value itself.
Very hard to predict.

The Iterative Imputer: Accuracy at a cost



Why use it?

- Preserves statistical relationships between variables.
- More accurate than simple mean imputation.

The Trade-off

- Computationally expensive (multiple regression loops).
- Must deploy the full training set to production for future predictions.

The Scenario: A sparse startup dataset

	R&D Spend	Administration	Marketing Spend
0	10	15	20
1	NaN	12	24
2	8	NaN	30
3	5	10	NaN
4	14	8	39

Target column (Profit) removed. Focus is on imputing input features.

Initialization: Filling the gaps with the Mean

	R&D Spend	Administration	Marketing Spend
0	10	15	20
1	9.25	12	24
2	8	11.25	30
3	5	10	28.25
4	14	8	39

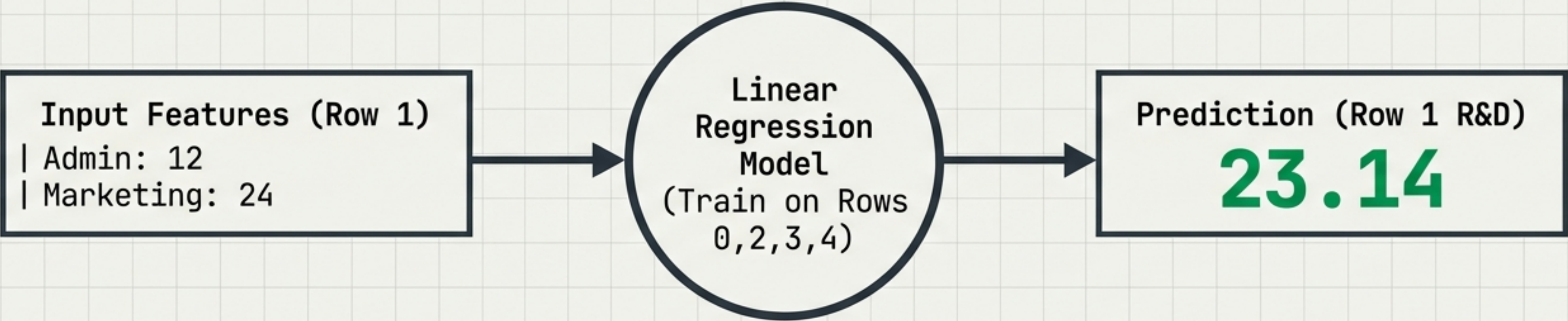
Col Mean: $(10+8+5+14)/4 = 9.25$

The First Chain: Targeting Column 1 (R&D)

Training Y	Features X			Prediction Input
	R&D Spend	Administration	Marketing Spend	
	0	10	15	20
	1		12	24
	2	8	11.25	30
	3	5	10	28.25
4	14	8	39	

Step 2.1: Remove the mean from the target column. Use other columns as features.

Iteration 1.1: The Linear Regression



	R&D Spend	Administration	Marketing Spend
0	23.14	12	24
1		12	24
2	18	12	24
3	20	12	20
4	23	12	25

The model predicts 23.14 based on the relationship between Admin and Marketing.

The Chain Reaction: Moving to Column 2

Features X			
	R&D Spend	Administration	Marketing Spend
0	10	15	20
1	23.14	11.06	24
2	8	11.25	30
3	5	10	28.25
4	14	8	39

Crucially, we use the NEW predicted value of R&D (23.14) to help predict Administration.

Completing the Cycle: Column 3

	Features X	Features X	Target Y
	R&D Spend	Administration	Marketing Spend
0	10	15	20
1	23.14	11.06	24
2	8	11.25	30
3	5	10	31.56
4	14	8	39

End of Iteration 1. Every missing value has been predicted once.

Calculating the Delta: Are we there yet?

Iteration 1 Prediction

23.14
11.06
31.56

—

Iteration 0 Mean

9.25
11.25
28.25

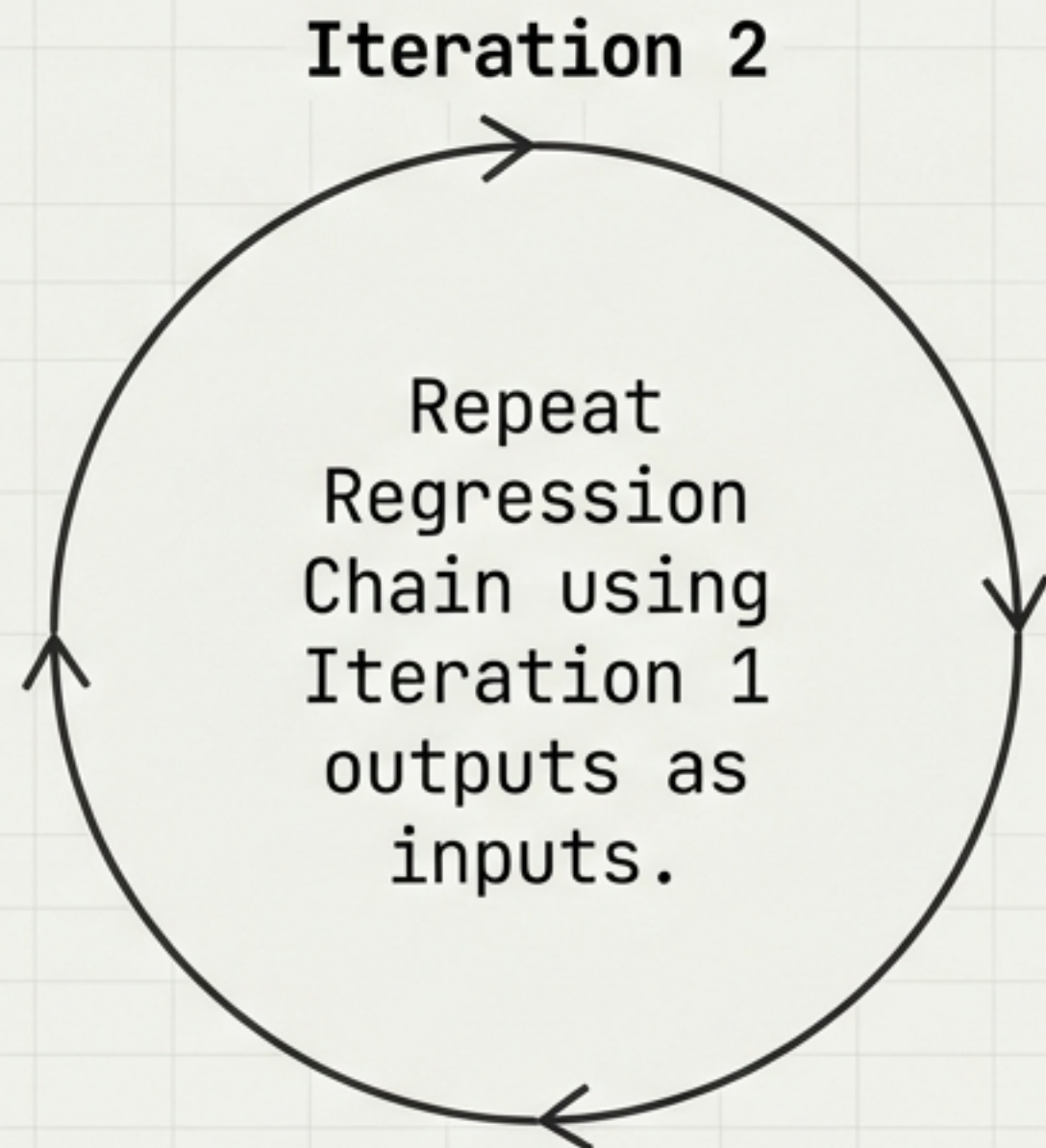
=

Delta / Difference

13.89
-0.19
3.31

Large differences (especially 13.89) indicate instability. The initial means were bad guesses, corrupting the first round of predictions.

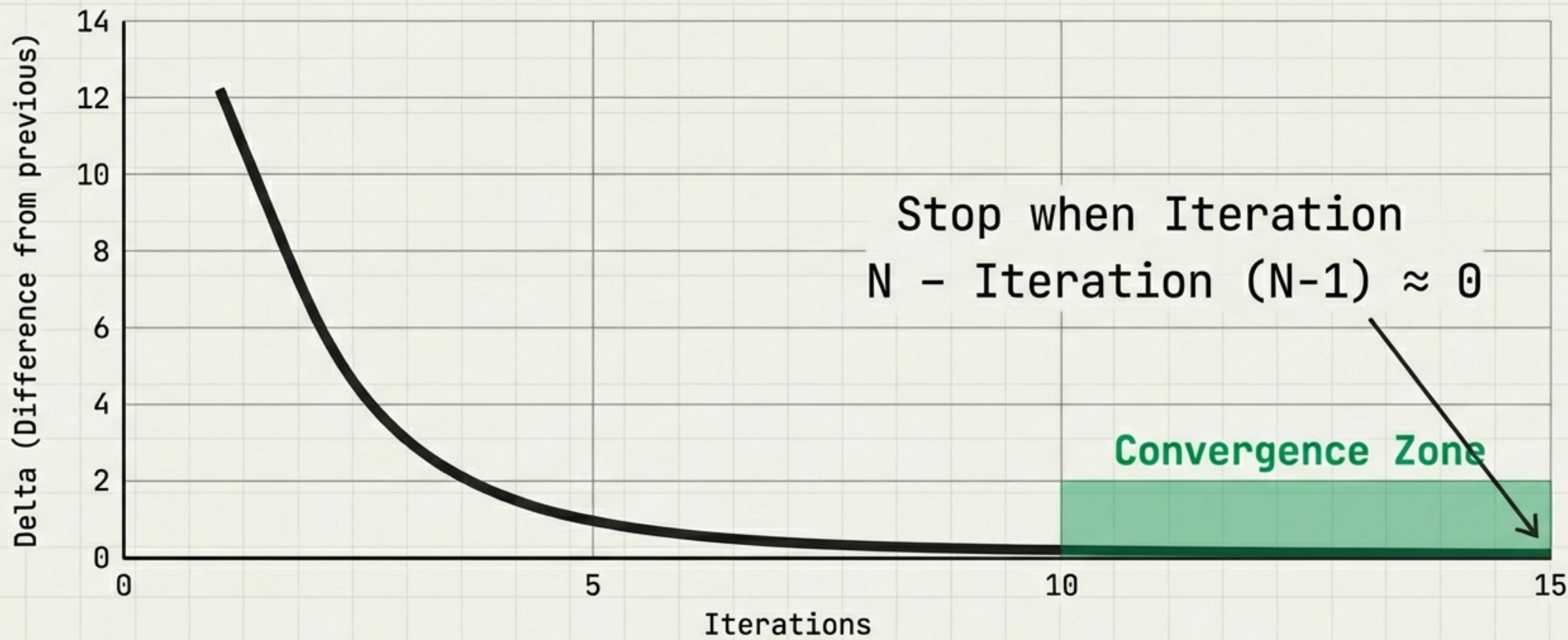
Iteration 2: Refining the Guess



	Features X		Target Y
	R&D Spend	Administration	Marketing Spend
0	10	15	20
1	23.78	11.22	24
2	8	11.25	30
3	5	10	31.56
4	14	8	39

The input data is cleaner, so the model sees a clearer signal.

Convergence: The Stopping Point



Usually stabilizes within 10-20 loops.

Python Implementation

```
from sklearn.impute import IterativeImputer  
from sklearn.linear_model import LinearRegression
```

```
lr = LinearRegression()  
imp = IterativeImputer(estimator=lr,  
                        max_iter=10,  
                        tol=1e-3)
```

```
imp.fit_transform(X)
```

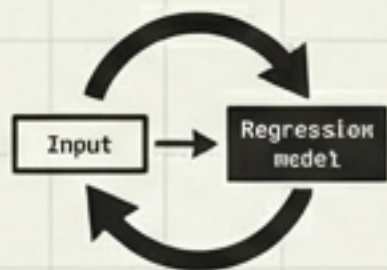
The model used for prediction
(Linear Regression, BayesianRidge, etc).

Number of cycles to run.

Summary & Best Practices



1. **Assumption:** Works best on data that is Missing At Random (MAR).



2. **Mechanism:** A cycle of regressions where output becomes input.



3. **Performance:** Slower than Mean Imputation, but yields **significantly better correlations**.



4. **Convergence:** Run for **10+ iterations** to ensure the values stabilize.

Thank You.

Generated by [Your Name/Model Name] /
Based on MICE Algorithm documentation.

